Rendeiro 2020 CLL — analysis summary

RUN 1 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=10) integrates all 7 samples: same-sample neighbour fraction 0.97 → 0.22, iLISI median 1.00 → 4.95 of 7. 205 cells in 3 detached islands removed; one connected component. Read on the regenerated embedding + per-neighbourhood iLISI, not a scalar: samples intermix AND the manifold is continuous.

Gate 2 — module geography → GO (non-branching)

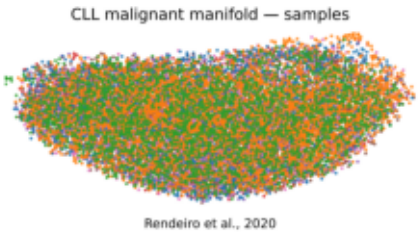
Non-branching case: a single (Gast)→Crest developmental gradient. Gast opposes Crest in space (smoothed r=−0.75; pSom weakly opposes both). Along pseudotime Gast falls (ρ −0.64) while Crest rises to a terminal tip (Crest-high decile pt=0.74) and pSom rises (ρ +0.46). Joint Gast&Crest enrichment 0.27 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — hub cross-repression → NA

Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks; a PPT tree throws 6–7 spurious forks on this near-round cloud). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks not produced, per spec).

STEP FIGURES

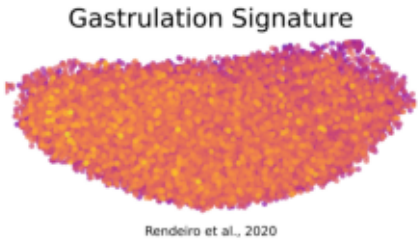
Manifold — samples



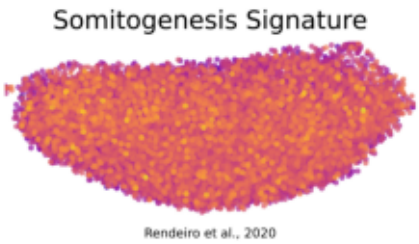
Pseudotime



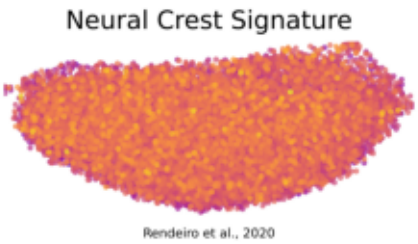
Gastrulation



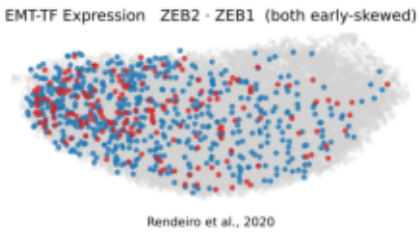
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
CLL peripheral blood (Rendeiro et al., 2020), 10x — 18,452 malignant cells over 4 patients (7 longitudinal samples); single diffuse-cycling body (~44% cycling, cell-cycle-free axis).

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation → neural-crest axis, Crest terminal. Gate 1 GO / Gate 2 GO / Gate 3 NA. CLL barely expresses EMT-TFs — only ZEB2 and ZEB1 clear the detection floor, both root-skewed (no terminus EMT-TF).

Parameter choices

Sample threshold	geom-mean 284 (cliff 1012→80, ratio 12.6) → 7/11 samples; 18,452 cells	**Overdispersed genes**	528 (scFates find_overdispersed)
Integration	Harmony on sample, θ=10 (mixing plateau; patient-private CLL batch)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 205 cells / 3 islands removed
Geometry	Non-branching scFates curve N=20 λ=0.3 μ=100; 2 tips/0 forks, SA 2.58; corr(x_lin,t)=0.92	**Root**	tip 0 = Gast/pSom-high, Crest-low origin
Module means (raw)	Gast −0.020 · pSom +0.008 · Crest +0.010 (Linear-pSom/Crest)	**Cell cycle**	43.8% cycling, diffuse single lobe; axis cell-cycle-free
EMT-TFs shown	ZEB2 + ZEB1, both root-skewed (no terminus EMT-TF); >3% gate		